Analyzing the Mass-Duality of the Koide Formula

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Abstract

This paper briefly overviews a preliminary result on the way from Aethic reasoning [1] to its proposed active reasoning followup [1]. The idea is meant to illustrate the relevance of what is termed “the principle of twos” by showing that ontological decomposition into two factors reproduces the Koide formula [2] over the leptons.

1 Stating the Principle of Twos

Broadly speaking, the principle of twos is an ontological prospect supposing that the presentation of quantum interference as a difference in actions in the Quantum Nexus [1]—as is itself effectively equivalent with the Born rule involving the power of two of its square magnitude—is itself merely one representative from a wider list of concepts wherein a system is necessarily decomposed into two elements.

Principle 1 (Principle of Twos)
Perhaps there is a single underlying physical-ontological circumstance which we should be targeting when attempting to bridge Aethic reasoning [1] with the Born rule's insistence [1] of a difference of actions in the Quantum Nexus, as opposed to only attempting to understand the action and the Born rule in isolation.

Here are a few examples of possible results which root to this same insistent-decomposability structure into two pieces.

1. *The action difference of two paths seen in the Quantum Nexus [1].*
2. *The Chiral decomposition of fermionic spinors.*
3. *The trigonometric factor decomposition of the lepton masses.*
   - *This third proposal will be articulated in the next section.*

More generally, perhaps we also interpret the “twoness” of particle physics as indicating a deeper ontological necessity. This paper does not articulate what this could be in particular, but the point of the matter is that it is worth considering a future theory for which this is structurally central.

2 Proposal for the Koide Formula

Let us begin this section by formulating an ontologically-motivated rendition of the Koide formula, that way we can solve the paradox of its existence by providing it with real ontological status in active reasoning. Here is the standard form of the Koide formula itself [2].

$$Q = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = 0.666661(7) \approx \frac{2}{3} \tag{1}$$

Now, our approach to motivating the Koide formula ontologically is centered on a core proposition: *within their genuine qualitative ontological framing, the three leptons themselves ought to fundamentally be cyclic rather than sequentially-ordered in character, which is to say that there ought to exist some transformation which uniquely maps a quantitative indicator of the electron mass to the muon mass, of the muon mass to the tauon mass, and of the tauon mass back to the electron mass. Under such a framing, then, we may understand the tauon not to be the “upper ceiling” in the first place,* *but instead merely a rung in a cyclically-constructed ladder.* More generally, the motivation of seeing things this way is so that we may properly enable the prospect of rendering the difference between any two leptons to simply a perspectival shift. For example, even in the case of a muon-tauon interaction, we would be apply to apply one rung of the aforementioned transformation in order to convert the tauon to an electron and the muon to another tauon. The local structure of spacetime may very well have to drastically change in order to accommodate this transition, but the fundamental idea is that the interaction in question was never "objectively" between a muon and a tauon in the first place—merely that was the joint effect of the underlying matter object and our convention of defining the surrounding spacetime.

With this ontological goal in mind, here is a derived solution to the Koide formula with the explicit aim of enabling such. We may start by considering the three square-root lepton masses themselves.

$$\sqrt{m_e}\,, \quad \sqrt{m_\mu}\,, \quad \sqrt{m_\tau} \tag{2}$$

Given these three quantities, now, let us label with with variables a, b, and c, although without yet knowing explicitly which label applies to which, (that is, so we may buy ourselves that much more flexibility in the coming algebra).

$$\{a, b, c\} \leftrightarrow \left\{\sqrt{m_e}\,, \sqrt{m_\mu}\,, \sqrt{m_\tau}\right\} \tag{3}$$

Now let us explicitly think to assume the following forms for each of a, b, and c, given two unknown angles α and β, and an arbitrary mass scale μ^2.

$$\begin{aligned} a &= \mu\cos(\alpha)\cos(\beta) \\ b &= \mu\cos\left(\alpha + \frac{\pi}{3}\right)\cos\left(\beta + \frac{\pi}{3}\right) \\ c &= \mu\cos\left(\alpha + \frac{2\pi}{3}\right)\cos\left(\beta + \frac{2\pi}{3}\right) \end{aligned} \tag{4}$$

Fundamentally speaking, then, where a^2, b^2, and c^2 represent the three lepton masses, but in an unknown order, then our desired "shifting" transformation can be enacted simply by phase-shifting α and β by 60° each.

$$f : \left(a^2, b^2, c^2\right) \mapsto \left(b^2, c^2, a^2\right) \quad \equiv \quad f : (\alpha, \beta) \mapsto \left(\alpha + \frac{\pi}{3}, \beta + \frac{\pi}{3}\right) \tag{5}$$

Note that a shifting constant of $\frac{\pi}{3}$ is sufficient, because a^2 is expressible as the product of two cosine squared functions, and cosine squared itself has a period of exactly π. As a result, we ought best to represent the shift as one third this interval, so that c^2 properly maps back onto a^2 when the cycle repeats. Regarding why the product of two cosines, each corresponding to an angle, was chosen, such was simply a practical intuition meant to provide us with a slight excess of flexibility prior to narrowing down our options to hopefully a single angle.

Indeed, is may be seen that such an approach is possible, which we may observe by substituting these three values into the Koide formula itself. That is, due to the commutative property of scalar addition, we understand that either of $a + b + c$ will equal $\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}$, and also of course that $a^2 + b^2 + c^2$ will equal $m_e + m_\mu + m_\tau$. Accordingly, then, we wish for the following to result, as per the statement of the Koide formula itself.

$$\frac{a^2 + b^2 + c^2}{(a + b + c)^2} = \frac{2}{3} \tag{6}$$

Indeed, we may now substitute our expressions for a, b, and c into either of the numerator and denominator terms.

$$\begin{aligned} a^2 + b^2 + c^2 &= m_0\cos^2(\alpha)\cos^2(\beta) + m_0\cos^2\left(\alpha + \frac{\pi}{3}\right)\cos^2\left(\beta + \frac{\pi}{3}\right) + m_0\cos^2\left(\alpha + \frac{2\pi}{3}\right)\cos^2\left(\beta + \frac{2\pi}{3}\right) \\ &= \frac{3m_0}{8}\left(2 + \cos(2(\alpha - \beta))\right) \end{aligned} \tag{7}$$

$$\begin{aligned} a + b + c &= \sqrt{m_0}\cos(\alpha)\cos(\beta) + \sqrt{m_0}\cos\left(\alpha + \frac{\pi}{3}\right)\cos\left(\beta + \frac{\pi}{3}\right) + \sqrt{m_0}\cos\left(\alpha + \frac{2\pi}{3}\right)\cos\left(\beta + \frac{2\pi}{3}\right) \\ &= \frac{3\sqrt{m_0}}{2}\cos(\alpha - \beta) \end{aligned} \tag{8}$$

We may now substitute these rather simplified terms into the Koide formula expression itself.

$$\frac{2}{3} = \frac{\frac{3m_0}{8}\left(2+\cos(2(\alpha-\beta))\right)}{\left(\frac{3\sqrt{m_0}}{2}\cos(\alpha-\beta)\right)^2} = \frac{1}{6}\left(2+\sec^2(\alpha-\beta)\right) \tag{9}$$

We may then solve for $\alpha - \beta$ directly, so as to use the Koide formula itself to allow only a single angular degree of freedom between them.

$$|\alpha-\beta| = \frac{\pi}{4} \tag{10}$$

Now, we know from empiricism the following identity, being all that is needed to fix the value of α itself, as relates to the mass ratio [3] of the muon to the electron.

$$\frac{c^2}{a^2} = \left(\frac{\cos\left(\alpha+\frac{2\pi}{3}\right)\cos\left(\beta+\frac{2\pi}{3}\right)}{\cos(\alpha)\cos(\beta)}\right)^2 = \frac{m_\mu}{m_e} \approx 206.768282708 \tag{11}$$

Importantly, note that in order to attain the correct three values for the lepton masses with a^2, b^2, and c^2 in the forms expressed above, it must follow that $a = \sqrt{m_e}$, $b = \sqrt{m_\tau}$, and $c = \sqrt{m_\mu}$. In other words, this is merely because the particular functions we chose happen to, in parity, transition in the "backwards" direction as we were anticipating for the leptons themselves as based on mass magnitude alone. The correction to this, then, is to allow c to be the square root muon mass, with a being fixed as the square root electron mass. All things considered, then, we attain the following value for α itself.

$$\begin{aligned} &\alpha \approx 1.59058499716 \approx 91.134^\circ \\ &\beta = \alpha + \frac{\pi}{4} \end{aligned} \tag{12}$$

Lastly, now, let us construct official formulae for the three lepton masses, as parametrized by the angular value θ_L, which we will refer to as "the lepton angle". We may then define such an angle as follows in terms of α itself.

$$\theta_L = \pi - 2\alpha \quad = \frac{3\pi}{2} - 2\beta \tag{13}$$

Note that this framing is exactly that which allows for the transformation $\theta_L \mapsto \theta_L + \frac{2\pi}{3}$ to precisely transition $m_e \to m_\mu \to m_\tau \to m_e \to \cdots$, being the exact conventionally-appealing order.

Upon expressing a, b, and c in terms of the parameter θ_L and a mass scale m_0, then, and we attain the following final expression for the three lepton masses, where u is the unified atomic mass unit. Note that these three formulae are entirely logically equivalent to the ones for a, b, and c which we had earlier, just in a drastically algebraically-simplified form. Please also note that m_0 is defined as $m_0 = \mu^2/4$ for our earlier mass scale μ^2.

$$\boxed{\begin{aligned} &\tilde{m}(\theta) = m_0\left(1-\cos(\theta)\right)\left(1-\sin(\theta)\right) \\ &\quad m_e = \tilde{m}(\theta_L) \\ &\quad m_\mu = \tilde{m}\left(\theta_L + \frac{2\pi}{3}\right) \\ &\quad m_\tau = \tilde{m}\left(\theta_L + \frac{4\pi}{3}\right) \\ &\quad m_0 \approx 1228.403306\ m_e \approx 0.67387737\ \text{u} \\ &\quad \theta_L \approx -0.039577341 \approx -2.267614588^\circ \end{aligned}} \tag{14}$$

Importantly, note that value of m_0 itself becomes entirely deducible by solving for it in any of the three equations for the lepton masses. Furthermore, the value of m_0 is also derivable as the average of the lepton masses.

$$m_0 = \frac{m_e + m_\mu + m_\tau}{3} \tag{15}$$

Highly intriguingly, we have due to our algebraic confirmation that these three lepton mass equations are collectively logically equivalent to the Koide formula itself, that is without even having to fix a value for m_0 or θ_L.

$$
\begin{aligned}
\text{For } f(\theta) = \frac{\sqrt{m_0}}{\sqrt{2}}(\cos(\theta)+\sin(\theta)-1)\,, \quad \tilde{m}(\theta) = (f(\theta))^2 = f^2(\theta)\,, \\
3\,m_0 = f^2(\theta) + f^2\left(\theta + \frac{2\pi}{3}\right) + f^2\left(\theta + \frac{4\pi}{3}\right), \\
-\frac{3}{\sqrt{2}}\sqrt{m_0} = f(\theta) + f\left(\theta + \frac{2\pi}{3}\right) + f\left(\theta + \frac{4\pi}{3}\right), \\
\frac{2}{3} = \frac{f^2(\theta) + f^2\left(\theta + \frac{2\pi}{3}\right) + f^2\left(\theta + \frac{4\pi}{3}\right)}{\left(f(\theta) + f\left(\theta + \frac{2\pi}{3}\right) + f\left(\theta + \frac{4\pi}{3}\right)\right)^2}
\end{aligned}
\tag{16}
$$

Indeed, the central symmetric statement about these equations are that upon mapping $\theta_L \mapsto \theta_L + 120°$, we get that the electron mass transitions to the muon mass, the muon mass to the tauon mass, and the tauon mass to the electron mass in cycle.

One last remark for this section is that, perhaps rather unexpectedly, we have the following almost exact approximation for this empirically-deduced value of θ_L, that is to an error of one part in ten million per radian.

$$
\theta_L \approx \frac{2}{9} - \frac{\pi}{12} \tag{17}
$$

This seems to suggest that the angle θ_L may itself very well be fixed to an extent, in which case the ratios between the lepton masses would no longer be free parameters at all—instead they would be exact. Is it at all possible that perhaps there is a Cauchy sequence representation of θ_L, for which this is an early term? Or perhaps this is the lepton angle's exact value already? If this value is indeed exact, then that gives us the following closed-form solutions for the three lepton mass ratios, (by substitution into their three mass expressions).

$$
\begin{aligned}
\frac{m_\mu}{m_e} &= \left(\frac{\frac{1}{\sqrt{2}} + \sin\left(\frac{2}{9} - \frac{\pi}{6}\right)}{\frac{1}{\sqrt{2}} - \sin\left(\frac{2}{9} + \frac{\pi}{6}\right)}\right)^2 \approx 206.77 \\
\frac{m_\tau}{m_e} &= \left(\frac{\frac{1}{\sqrt{2}} + \cos\left(\frac{2}{9}\right)}{\frac{1}{\sqrt{2}} - \sin\left(\frac{2}{9} + \frac{\pi}{6}\right)}\right)^2 \approx 3477.47 \\
\frac{m_\tau}{m_\mu} &= \left(\frac{\frac{1}{\sqrt{2}} + \cos\left(\frac{2}{9}\right)}{\frac{1}{\sqrt{2}} + \sin\left(\frac{2}{9} - \frac{\pi}{6}\right)}\right)^2 \approx 16.82
\end{aligned}
\tag{18}
$$

Regarding the masses individually, they could also be simply represented in such a case as follows.

$$
\begin{aligned}
m_e &= m_0\left(1 - \cos\left(\frac{2}{9} - \frac{\pi}{12}\right)\right)\left(1 - \sin\left(\frac{2}{9} - \frac{\pi}{12}\right)\right) \\
m_\mu &= m_0\left(1 - \cos\left(\frac{2}{9} + \frac{\pi}{12}\right)\right)\left(1 + \sin\left(\frac{2}{9} + \frac{\pi}{12}\right)\right) \\
m_\tau &= m_0\left(1 + \cos\left(\frac{2}{9} + \frac{\pi}{4}\right)\right)\left(1 + \sin\left(\frac{2}{9} + \frac{\pi}{4}\right)\right)
\end{aligned}
\tag{19}
$$

Such perhaps is interesting, to say the least.

3 Conclusion

Altogether, this illustration may be potentially helpful both for the pursuit of identifying a general circumstance behind the recurring two-part-decomposition in a number of structures throughout quantum field theory, and also for the ontological reframing of the lepton generations themselves as possibly involving a deeper transformation for which each individually is merely a relativist presentation. Onward now to the wider question of active reasoning [1].

References

[1] A. Benander, "Aethic reasoning: A comprehensive solution to the quantum measurement problem," November 2024.

[2] A. Rivero and A. Gsponer, "The strange formula of Dr. Koide," May 2005.

[3] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, "CODATA recommended values of the fundamental physical constants: 2022," *Rev. Mod. Phys.*, vol. 97, p. 025002, 2025.